

Manan Bhavsar

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Summary

Mechanical Design Engineer with 5 years of experience across rotating equipment, steam turbine components, marine propulsion systems, gearbox design, and power plant auxiliary systems. Delivered 1,050+ engineering drawings and models across programs for Wärtsilä, WinGD, and Siemens — operating within PLM-controlled environments (Teamcenter, COMOS) to international standards. Combines deep CAD expertise in Siemens NX with system-level engineering across piping, lube oil, compressor, and HVAC packages. Known for initiative — independently developed productivity frameworks that reduced drawing turnaround by ~50% on live programs.

Skills

Mechanical Engineering:

Rotating Equipment, Pumps, Compressors, Steam Turbines, Gearbox Design, Auxiliary Systems, HVAC Systems

Design & Analysis:

3D Modelling, Assembly Design, Manufacturing Drawings, GD&T, Tolerance Stack-up Analysis, Static & Modal FEA

EPC & Package Engineering:

Equipment Sizing, P&ID Development, Technical Bid Evaluation, Vendor Document Review, Datasheets, BOM/MTO Preparation

Software:

Siemens NX, Teamcenter, COMOS, AutoCAD, ANSYS, Microsoft Excel

Standards:

ASME B31.1, ASME B31.3, API Standards, ISO Standards, ASHRAE, ISHRAE

Experience

Mechanical Design Engineer | L&T Technology Services (Via Orcapod) | Vadodara

Jan 2026 – Present

- Rationalizing a ~30-drawing assembly package for a Wärtsilä marine propulsion system restructuring callout sequences.
- Building and fully constraining 3D assemblies in Siemens NX; resolving broken constraints, repairing corrupted references, and remodelling components where geometry recovery is not feasible.
- Executing environmental compliance material substitution (seals, O-rings, thread-locking compounds) per Wärtsilä-approved specifications with full revision traceability across drawings and MTO.
- Performing tolerance stack-up analysis in NX to validate assembly fit and function following component-level design changes.
- Preparing and updating manufacturing drawings, BOMs, and engineering documentation in Teamcenter while ensuring revision control compliance

Mechanical Engineer | Siemens Ltd. (Via Sphinx worldbiz Ltd.) | Vadodara

July 2024 – Dec 2025

- Delivered 270+ pre-steam piping weldment drawings across SST-300/400/500 turbine programs in Siemens NX — specifying weld sequences, machining instructions, and pipe fit-up geometry for turbine casing installation.
- Self-initiated and delivered a master model framework in NX where feature activation auto-updates linked drawings across upper/lower casing and base frame packages — reducing per-drawing turnaround by ~50%.
- Supported turbine package modeling — positioning PHEs, filters, and lube oil components on skids, performing clash checks and preparing package-level BOMs in Teamcenter.
- Developed ~80 control oil piping and turbine lifting arrangement drawings in AutoCAD with Excel-managed BOMs for procurement and installation teams.

Design Engineer | L&T Precision Engineering & Systems | Pune

Jan 2024 – June 2024

- Developed 1,050+ NX 3D models and manufacturing drawings across 25 sub-assemblies for a vehicle-mounted RADAR structure program — all parts from scratch through assembly-level documentation.
- Performed static and modal FEA in ANSYS, validating structural integrity.

Design Engineer | L&T Technology Services | Vadodara

June 2022 – Dec 2023

- Designed 5 centrifugal/screw compressor packages and 3 pump packages for Wärtsilä Gas Solutions (Finland) end-to-end equipment sizing, valve selection, P&ID; development in COMOS, datasheets, and full 3D skid models in NX.
- Designed marine lube oil, fuel, water, and drain piping systems for WinGD — studying reference designs, implementing reliability improvements, and delivering new systems with full 3D models and BOMs.
- Migrated thousands of legacy IDEAS parts into Siemens NX, establishing a usable 3D asset base for ongoing design workflows.

Trainee Engineer | Adept Consultants & Engineers | Vadodara

June 2022 – Dec 2023

- Designed complete HVAC systems for pharmaceutical clients — chilled water, VRF, and associated piping — covering equipment sizing, layout, and full documentation per ASHRAE/ISHRAE standards.
- Engineered compressor and pump packages for 3 clients — equipment sizing, technical bid evaluation, P&ID; development and complete BOM/MTO/BOQ procurement packages.

Education

BE Mechanical | 20217 -2021 | Gujarat Technological university